

SHUBHAM JAIN

Bengaluru, KA, India | shubhamajay2@gmail.com | +91 7096211133 | linkedin.com/in/shubham-jain

PROFESSIONAL SUMMARY

Data Scientist with hands-on production experience across machine learning, optimization, and GenAI/agent system delivery. Builds predictive models, optimization engines, and multi-agent AI workflows that move from notebook to production, spanning industrial manufacturing, insurance, and digital health use cases. Proficient in Python, SQL, PySpark, LLM orchestration (LangChain), RAG pipelines, and prompt engineering, with a track record of translating complex model outputs into stakeholder-ready insights and client-facing delivery (Aditya Birla Group, Fractal Analytics).

CORE SKILLS

Languages & Data: Python (Pandas, NumPy, Scikit-learn, XGBoost), SQL, PySpark, PostgreSQL

Machine Learning: Predictive Modeling, Time Series Forecasting, Feature Engineering, Causal Inference (DoubleML), Explainability (SHAP), A/B Testing & Experiment Design, Model Deployment

GenAI & Agentic AI: Multi-Agent Orchestration, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Evaluation & Observability, Responsible AI, LangChain, Semantic Search

Optimization: Mathematical Programming, Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Process & Constraint Optimization

Delivery & Tools: Databricks (PySpark-optimised), Power BI, Matplotlib, Seaborn, PyTorch, AWS, Azure DevOps (Git), MLOps Governance, Stakeholder Reporting & Documentation

PROFESSIONAL EXPERIENCE

Assistant Manager, Data Scientist | Aditya Birla Management Corporation Jan 2025 – Present | Bengaluru, KA

- Architected and deployed a production multi-agent AI system orchestrating optimization engines, predictive models, RAG pipelines, and SQL query agents — using prompt engineering, retrieval grounding, and deterministic routing for explainable agent behavior, surfaced as a no-code stakeholder dashboard that eliminated manual analytical intervention.
- Designed, trained, and deployed an LLM-XGBoost hybrid model for insurance claims fraud detection end-to-end, owning the full ML lifecycle from feature engineering and model evaluation to deployment — achieving an 18% accuracy uplift and 40% reduction in manual processing time.
- Built a spatiotemporal ML model for early disease hotspot detection through extensive EDA and feature engineering on geospatial and time-series data, delivering multi-week risk forecasts that generated Rs. 5M+ in business value.
- Led the analytics layer for a digital twin of calcination processes, integrating high-frequency sensor data with ML models for plant-wide decision support and simulation-based what-if analysis across production, quality, and energy trade-offs to improve first-time-right rates.
- Architected an optimization engine for cement manufacturing using mathematical programming and heuristic methods (GA/PSO), achieving a 10% reduction in energy consumption through enhanced process control and material flow optimization.
- Architected scalable data pipelines on Databricks optimised with PySpark, processing high-frequency sensor data from 4+ manufacturing lines via OPC UA into a centralized PostgreSQL database; managed CI workflows on Azure DevOps following MLOps best practices.
- Developed interactive MIS dashboards (Power BI, Matplotlib, Seaborn) to communicate model performance and data insights to senior stakeholders; maintained documentation of pipelines, model assumptions, and analytical findings.

Project Intern | Fractal Analytics

May 2024 – Jul 2024 | Bengaluru, KA

- Partnered directly with client stakeholders to translate B2B campaign data into predictive models and clustering solutions, driving a 10% lift in lead conversion for a high-value cohort.

- Designed and executed A/B tests to identify high-impact customer segments and optimize engagement strategies across multiple touchpoints.
- Translated complex model outputs into executive-ready insights and dashboards that directly shaped go-to-market decisions for senior leadership.

Quant Analyst Intern | Subhkam Ventures

Jun 2023 – Jul 2023 | Mumbai, MH

- Formulated quantitative trading strategies using moving averages and momentum signals, achieving 15% outperformance against NIFTY50 benchmarks.
- Deployed portfolio optimisation models across 1,100+ cash stocks, delivering a 25% increase in risk-adjusted returns.

EDUCATION

B.Tech, Data Science, Economics & Business Plaksha University | Aug 2021 – Jul 2025 | CGPA: 9.0 / 10

Coursework: Econometrics, Machine Learning, Advanced Statistics, Behavioral Economics, Game Theory, Marketing Analytics

CERTIFICATIONS & ACHIEVEMENTS

- Databricks ML Associate (Jan 2026)
- Speechcrafter — Toastmasters International (2025)
- VMart Undergraduate Scholarship — Top 1% of cohort
- HackPlaksha 2023 Winner — AI-Empathy Analytics Dashboard